Raspberry Pi audio boards overview

Source: https://github.com/raspberrypi/documentation/blob/7376533ab86ea07acc77ef6a9f3c3e57fcff833d/documentation/asciidoc/accessories/audio/introduction.adoc (CC BY-SA 4.0; adapted for this benchmark)

Raspberry Pi audio boards overview

Overview

Raspberry Pi Audio Boards bring high quality audio to your existing hi-fi or Raspberry Pi-based equipment and projects. We offer four different Hardware Attached on Top (HAT) options that will fit any Raspberry Pi using the 40-pin GPIO header.

Each board has a specific purpose and set of features. The highest audio quality playback is available from our DAC PRO, DAC+ and DigiAMP+ boards, which support up to full HD audio (192 kHz); while the Codec Zero supports up to HD audio (96 kHz) and includes a built-in microphone, making it ideal for compact projects.

Features at a glance

| Line out | Balanced out | Stereo speakers | Mono speaker | Headphones | Aux in | Aux out | Ext mic | Built-in mic

DAC Pro ^| ✓ ^| ✓ | | ^| ✓ | | | |

DAC+ ^| ✓ | | | ^| ✓ | | | |

DigiAmp+ | | ^| ✓ | | | | | |

Codec Zero | | | ^| ✓ | ^| ✓ ^| ✓ ^| ✓ ^| ✓

Line out:: A double phono/RCA connector, normally red and white in colour. This output is a variable

analogue signal (0-2V RMS) and can connect to your existing hi-fi (pre-amp or amplifier), or can be used

to drive active speakers which have their own amplifier built in.

Balanced out:: An XLR connector, normally a three-pin male connector. This is used in a studio set-up, and in some

high-end hi-fi systems. It can also be used to drive active monitor speakers like those used at clubs or on

stage directed towards the DJ or performers.

Stereo speakers:: Two sets of screw terminals for 2 × 25 W speakers. These are for traditional hi-fi speakers without built-in amplification. These are known as passive speakers.

Mono speaker:: A screw terminal for a single 1.2 W speaker, as found in a transistor radio or similar.

Headphones:: A 3.5 mm jack socket delivering stereo audio for a set of headphones. The headphone amplifiers on the Raspberry Pi DAC boards can drive up to 80/90 Ω impedance headphones.

Aux in:: A double Phono/RCA connector or 3.5 mm socket. Accepts analogue audio in up to 1V RMS. This can be used to record audio from a variable analogue source such as a mobile phone, MP3 player or similar.

Aux out:: A double Phono/RCA connector or 3.5 mm socket. Delivers analogue audio out up to 1V RMS. This can be used to feed audio into an amplifier at a reduced volume compared to Line out.

Ext mic:: A 3.5 mm socket for use with an external electret microphone. The built-in MEMS microphone on the Codec Zero is automatically disabled when the external Mic in connector is used.